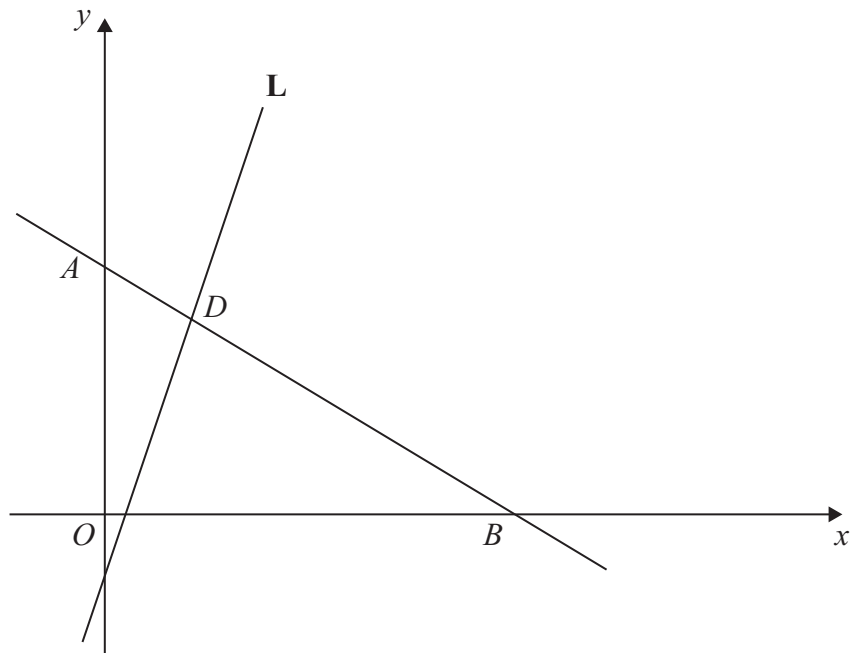




In the diagram

A is the point $(0, 8)$

B is the point $(16, 0)$

The point D divides the line segment AB in the ratio $1 : 3$

The line L passes through D .

The gradient of L is $\sqrt{3}$

L passes through the point with coordinates $(-2, f)$

Show that $f < -4$